Empirical Validation on a Humanoid Digital Twin

Draft subsection VII.H — Decentralised Capability Abstraction paper (V6.1)

G1 simulation campaign 2026-07-04 · 83 runs, 11 arms · pending review (Adrián / Renxi) — not yet inserted in the manuscript

The analytical validation above is complemented by an empirical campaign on a digital twin of the physical deployment: a Unitree G1 humanoid navigating a door-crossing delivery task (A↔B, ≈13 m, a 0.8 m doorway and a 0.71 m corridor pinch) in a Gazebo replica of the real laboratory, generated from the robot's own SLAM maps. The full production navigation stack (static-map global planner, local DWA, door-engagement controller) runs unmodified; the DCE runtime (Meta-Reasoner 2.0) governs it through the shared-experience bridge at 2 Hz, exactly as deployed on the physical robot.

The governance task is the payload condition of the delivery scenario: the robot carries an open cup of water. Spilling is modelled physically — first-mode liquid sloshing in a cylindrical cup (natural frequency $\omega_0 \approx 21$ rad/s for $R = 4$ cm), driven by the cup's horizontal acceleration including arm-offset and gait-bounce terms, with spills generated as a non-homogeneous Poisson process on surface elevation against freeboard. Consistently with the simulation study that motivated this design, jerk — not speed alone — dominates spill risk. A sealed lid multiplies the hazard rate by 0.25.

Governance arms ($N = 8$ runs each; lid arms $N = 6$; mean ± 95% CI) instantiate the architectural comparisons of Sections IV–VII:

Arm / governance	Reached	Time (s)	Spills/run	Risk (% time)
M0 baseline — none (0.30 m/s)	8/8	83.4 ± 5.2	0.50 ± 0.37	8.1 ± 1.4
M1 fixed-conservative — none (0.18 m/s cap)	8/8	110.8 ± 2.4	0.00	0.8 ± 0.2
Single-candidate governed (DCE, one analogy)	5/8 (3 HELP)	106.2 ± 5.1	0.12 ± 0.24	0.5 ± 0.2
M2 payload (proposed) — DCE, payload task	8/8	107.9 ± 4.2	0.00	0.4 ± 0.2
M2 + task-blind prior (miscalibrated task)	8/8	108.2 ± 3.5	0.00	0.7 ± 0.2
M0, sealed lid — none	6/6	88.1 ± 4.9	0.17 ± 0.33	10.6 ± 2.9
M2, sealed lid (twin-calibrated, covered-delivery)	5/6 (1 HELP)	102.5 ± 2.2	0.17 ± 0.33	0.7 ± 0.5

Table H-1. Digital-twin payload campaign. "Risk" is the fraction of run time with the liquid surface above 70% of freeboard. HELP = governed abort (sustained empty deployable set).

Findings

(1) Deployability governance converts spills into bounded time cost. The ungoverned baseline reaches the goal fastest but spills on half of its runs, spending 8.1% of each run with the liquid surface above 70% of freeboard. Active DCE governance under the payload task eliminates spills entirely (0/8 runs; risk 0.4%) at a 29% time cost, while preserving 8/8 task completion. The fixed-conservative baseline achieves the same safety by construction, but rigidly: it cannot re-certify a faster capability when conditions permit (finding 4).

(2) HELP is the correct outcome of an empty deployable set. When the candidate set is reduced to a single analogy and a hard-veto meta-parameter (mobility, the resistance channel) enters its dangerous region while the robot presses the corridor pinch, no deployable candidate remains; the runtime returns sustained HELP and the experience-escalation layer aborts the run (3/8 runs). This is not a failure of the mechanism but its specified semantics: capability-validity governance must refuse to certify rather than silently persist (conservative non-selection).

(3) The calibration-limited regime is observable in deployment. Running the physical laboratory's QoE calibration unchanged on the digital twin — whose clearance distribution is systematically narrower (cruise median 0.76 vs ≈1.0) — left the efficient capability non-certifiable for entire runs: an instance of the θ_QoE calibration-error budget of Table IV arising live rather than analytically. Re-calibrating the QoE boundaries from measured twin distributions (per the interface-refinement pattern) restored certification in open stretches without touching the reasoner or the physical-robot configuration.

(4) Task-conditioned analogy selection exploits payload changes. With a sealed lid, the ungoverned baseline improves only passively (spills 0.50 → 0.17/run through physics; its risk exposure is unchanged at 10.6%). The governed system, given the covered-delivery task configuration on the calibrated twin, actively exploits the change: 102.5 ± 2.2 s versus 111.5 ± 7.3 s for the same governance under the open-cup configuration — a policy-level gain unavailable to fixed baselines, consistent with the lid-state delta of the 2-D study.

(5) Temporal scales of correction separate as designed. A task-blind (wrong) prior is corrected within runs in ≈4 s: per-tick shared-experience evidence (clearance) strips the miscalibrated capability's certification before its first failure, so the cross-run analogical-trust layer receives no failure signal to learn from on the twin. The cross-run layer — per-analogy Dempster–Shafer belief over {match, mismatch}, persisted between runs, with plausibility-gated deployment and a conservative policy blend — is validated mechanistically (plausibility trace 1.00 → 0.75 → 0.53 → 0.36 over three spill runs, with corrected zero-shot start on the fourth). Its operational regime is the physical laboratory, where cruise clearance certifies the fast capability and gait-induced spills are invisible to per-tick channels: exactly the failure mode the run-level trust layer exists to catch. The physical-robot campaign (H.1) exercises it.

H.1 Physical-robot protocol (in progress)

Twenty runs on the physical G1 in the mapped laboratory, water-cup payload, four conditions (baseline / governed × open / sealed lid), spill ground truth marked manually per event (UDP channel, timestamped into the per-tick dataset), Layer-2 belief state persisted across runs (G1_M2_STATE). Metrics identical to the twin campaign; runs tagged env=real in the shared results table.

Annex — Internal notes (not for the manuscript)

The zero-spill outcome of the wrong/wrongdst/wrongdstsim arms is NOT an experimental failure: it is finding (5). If the cross-run recovery curve is wanted IN SIMULATION, the protocol faithful to the thesis (Ch. 5, 5.3.6.3) is to force the wrong analogy (lock) for the first K runs — implementable as G1_M2_LOCK — but the natural place to capture it is the physical robot, where open-floor cruising genuinely certifies the Efficient capability.

The table omits the redundant arms (wrongsim / wrongdst / wrongdstsim); they are available via analyze_campaign.py and in runs_summary.csv (filter env=sim, sim_id lab_v1_payload_*).

Contrast with the thesis 2-D simulation (warehouse-cluttered): M0 SE=13.0 / 2.12 spills·run; M2 SE=37.6 / 0.28. The 3-D twin reproduces the same safety ordering M0 < M1 ≈ M2 with physically grounded sloshing.

Reproduction: python3 sim_campaign.py (arms via CAMPAIGN_ARMS) · python3 analyze_campaign.py · commits up to 0b6ea24 on main.